

Program : Diploma in Robotic Process Automation	
Course Code : 4301	Course Title: Web Scripting Languages
Semester : 4	Credits: 0
Course Category: Program Core	
Periods per week: 3 (L: 3 T: 0 P: 0)	Periods per semester: 45

Course Objectives:

- To understand the principles of creating an effective web page.
- To understand the concepts of JavaScript.
- To implement client side interface through DOM, JQuery & AJAX.

Course Prerequisites:

Topic	Course code	Course name	Semester
Basic programming skills in C language.		Introduction to Computation and Programming	2
ICT lab			1

Course Outcomes:

On completion of the course student will be able to:

CO_n	Description	Duration (Hours)	Cognitive Level
CO1	Illustrate the basic concepts of Internet, world wide web, client server architecture and formulation of web pages using HTML.	10	Understand
CO2	Illustrate the concepts of JavaScript and validate web pages using JavaScript.	11	Understand
CO3	Apply DOM to add functionality to web pages.	10	Applying.
CO4	Use Ajax, JQuery & AngularJS to enhance the functionality of web pages..	12	Applying.
	Series Test	2	

CO – PO Mapping:

Course Outcomes	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7
CO1	3				3	3	
CO2	3				3	3	
CO3	3				3	3	
CO4	3			3	3	3	3

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline

	Description	Duration (Hours)	Cognitive Level
CO1	Illustrate the basic concepts of Internet, world wide web, client server architecture and formulation of web pages using HTML.		
M1.01	Define Internet and world wide web, web browser and world web addresses (IP, Domain, URL),	1	Understanding
M1.02	Summarize Client server architecture, web browser and web server communication.	1	Understanding
M1.03	Define XML, Summarize XML with example.	1	Understanding
M1.04	Define HTML, Show basic structure of HTML, Summarize head, title and metadata in HTML Summarize heading tags in HTML.	1	Understanding
M1.05	Summarize paragraph and line break tags in HTML, Summarize Lists in HTML, Summarize Tables in HTML	1	Understanding
M1.06	Summarize Image tag and its attributes in HTML, Summarize hyperlinks in HTML, Summarize Form tag HTML.	2	Understanding
M1.07	Summarize Form elements in HTML (Text boxes, Radio Buttons, Check Boxes, Text Area and Buttons.)	1	Understanding
M1.08	Develop a web page using all the above-mentioned HTML elements.	2	Applying
Contents: Define Internet - World Wide Web - Web Browser-Web Server and Web Addresses (IP, Domain, URL) -Client Server Architecture-Web Browser and Web Sever communication, XML. Describe HTML, Basic structure of HTML page, Basic Mark-up: -Headings - Paragraphs - line breaks –Lists – Tables-Image Tag and Attributes– Hyperlinks-Form, Form elements in HTML: - Text boxes, Radio Buttons, Check Boxes, Text Area and Buttons.			
CO2:	Illustrate the concepts of JavaScript and validate web pages using JavaScript.		
M2.01	Define JavaScript, How JavaScript makes the webpages more interactive, examples of JavaScript in browser.	1	Understanding
M2.02	Basic JavaScript instructions: statements, comments, variable, data types.	1	Understanding
M2.03	JavaScript arrays, expressions, operators.	1	Understanding

M2.04	Functions methods: function, anonymous function, variable scope.	2	Understanding
M2.05	Functions objects: object, this, arrays are objects, browser object model, document object model.	2	Understanding
M2.06	Global objects: string, number, math, date.	1	Understanding
M2.07	Decision making and Loops: decision making: if statement, if...else statement, switch statement, loops: key loop concepts, for loops, while loops, do while loops.	3	Applying
	Series Test – I	1	

Contents:

Define JavaScript – How JavaScript makes the webpages more interactive - examples of JavaScript in browser, Basic JavaScript instructions: statements, comments, variable, data types, arrays, expressions, operators. Functions methods: function, anonymous function, variable scope, object, this, arrays are objects, browser object model, document object model, Global objects: string, number, math, date. Decision making and Loops: decision making: if statement, if...else statement, switch statement, loops: key loop concepts, for loops, while loops, do while loops.

CO3:	Apply DOM to add functionality to web pages.		
M3.01	DOM:Introduction to Document Object Model (DOM), DOM tree as a model of a web page.	2	Understanding
M3.02	DOM tree - working with DOM tree accessing elements, nodelists, selecting elements: using class attribute, tag name.	2	Understanding
M3.03	DOM - Traversing, adding or removing html content, update text and markup, adding/removing elements.	2	Applying.
M3.04	Event handling: different event types, three ways to bind an event to an element, using DOM event handlers.	2	Applying.
M3.05	Describe event listeners, using parameters with event listeners, the event object, event delegation, user interface events, event bubbling.	2	Applying.

Contents: DOM: Introduction to Document Object Model (DOM), DOM tree as a model of a web page, working with DOM tree, accessing elements, nodelists, selecting elements: using class attribute, tag name. DOM - Traversing, adding or removing html content, update text and markup, adding/removing elements. Event handling: different event types, three ways to bind an event to an element, using DOM event handlers. Describe event listeners, using parameters with event listeners, the event object, event delegation, user interface events, event bubbling.

CO4:	Use Ajax, JQuery & Angular JS to enhance the functionality of web pages.		
M4.01	AJAX: What is Ajax?, Why use Ajax?, How Ajax works?	1	Understanding
M4.02	Handling Ajax request and response, data format: XML.	1	Applying.
M4.03	JSON – Introduction to JSON, Working with JSON data, Loading HTML & JSON with Ajax.	2	Applying
M4.04	JQUERY : What is JQuery ?, A basic JQuery example, Why use JQuery ?	2	Understanding
M4.05	JQUERY Operations - finding elements, JQuery selection, getting element content, updating elements, changing content, inserting elements, adding new content, getting and setting attributes.	2	Applying
M4.06	Angular JS: What is Angular JS?, Why use Angular JS?, How Angular JS works?.	1	Understanding
M4.07	Angular Expression, Filters, Directives, Controllers.	1	Understanding
M4.08	Familiarization of Angular JS modules & Forms.	2	Applying
	Series Test – II	1	

Contents:

Introduction to AJAX, Handling Ajax request and response, data format: XML, Introduction to JSON, Working with JSON data, Loading HTML & JSON with Ajax, Introduction to JQUERY, JQUERY Operations - finding elements, JQuery selection, getting element content, updating elements, changing content, inserting elements, adding new content, getting and setting attributes. Introduction to Angular JS, Angular Expression, Filters,

Directives, Controllers. Angular JS modules & Forms.(no detailed study required.)

Text / Reference

T/R	Book Title/Author
T1	Paul J. Deitel, Harvey M. Deitel, Abbey Deitel, “ Internet and World Wide Web How To Program ”, 5/E, Pearson Education, 2012.
R1	Jon Duckett , “ HTML and CSS: Design and Build Websites ”, Wiley
R2	Jon Duckett , “ JavaScript and JQuery : Interactive Front-End Web Development ”, Wiley.
R3	Felipe Coury, Ari Lerner, Carlos Taborda” ng – book: The Complete Guide to Angular Paperback - 6” , February 2018.

Online Resources

Sl.No	Website Link
1	https://www.w3schools.com/html/default.asp
2	https://www.tutorialspoint.com/html5/index.htm
3	https://www.w3schools.com/js/default.asp
4	https://www.w3schools.com/angular